

Homoscedasticity and Its Testing: From Conditional Expectation to the Breusch–Pagan Test

Boris Garbuzov

Part 12 / 12

**The Breusch–Pagan Test:
Likelihood Foundations and Score Analysis**

July 2026

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1 The Breusch–Pagan Test: Likelihood Foundations and Score Analysis

1.1 Where the likelihood comes from

We follow Boris’s notes of July 5, 2026 (note 58).

The question

Boris interrupted the Probabilist’s exposition of the score and asked: where does the likelihood come from in the first place? The Probabilist proposed defining the likelihood through ε . Boris objected: if $\varepsilon \sim \mathcal{N}(0, 1)$ and its distribution does not depend on β , then β will not appear in the likelihood. The Probabilist replied: the distribution of ε does not depend on β , but the *observed value* of ε does. This distinction is the key.

The model

We work in the simplest case: $p = 1$, $z = x$, intercept absorbed. The full model is

$$\begin{aligned} Y &= \beta_0 + \beta_1 X + U, \\ U &= \sqrt{h(\alpha_0 + \alpha_1 X)} \cdot \varepsilon, \\ \varepsilon &\perp X, \quad \varepsilon \sim \mathcal{N}(0, 1). \end{aligned}$$

Introducing the shorthand $\sigma_x^2 := h(\alpha_0 + \alpha_1 x)$, this becomes

$$Y = \beta_0 + \beta_1 X + \sigma_X \cdot \varepsilon.$$

What is given (observed): X, Y . What is deterministic and unknown: $\beta_0, \beta_1, \alpha_0, \alpha_1$ (and h , treated as known in BP). What is random and unobserved: U, ε , and — in the random-regressor framework — X itself. The distribution of X is a nuisance: it factors out of the conditional likelihood.

Expressing ε through observables

Inverting the model for ε :

$$\varepsilon = \frac{Y - \beta_0 - \beta_1 X}{\sqrt{h(\alpha_0 + \alpha_1 X)}} = \frac{Y - \beta_0 - \beta_1 X}{\sigma_X}.$$

This expression is not observed directly — it depends on the data (X, Y) *and* on the parameters (β, α) . Call it $\varepsilon(y, x; \beta, \alpha)$ to make the dependence explicit. For the i -th observation:

$$\varepsilon_i(\beta, \alpha) := \frac{y_i - \beta_0 - \beta_1 x_i}{\sigma_{x_i}}, \quad \sigma_{x_i}^2 = h(\alpha_0 + \alpha_1 x_i).$$

Why β participates in the likelihood

Boris’s worry was: $\varepsilon \sim \mathcal{N}(0, 1)$ with no parameters in its distribution, so β cannot appear. The resolution: the distribution of ε is parameter-free, but the *value* it takes at observation i is $\varepsilon_i(\beta, \alpha)$, which depends on β . The likelihood is the density of the observed data evaluated at the observed values — and the transformation from (y_i, x_i) to (ε_i, x_i) has a Jacobian.

Concretely: since $\varepsilon_i = (y_i - \beta_0 - \beta_1 x_i)/\sigma_{x_i}$, the map $y_i \mapsto \varepsilon_i$ has Jacobian $\partial \varepsilon_i / \partial y_i = 1/\sigma_{x_i}$. By the change-of-variables formula, the density of Y_i given $X_i = x_i$ is

$$p(y_i | x_i; \beta, \alpha) = \phi(\varepsilon_i(\beta, \alpha)) \cdot \frac{1}{\sigma_{x_i}},$$

where ϕ is the standard normal density. Writing this out:

$$p(y_i | x_i; \beta, \alpha) = \frac{1}{\sqrt{2\pi} \sigma_{x_i}} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma_{x_i}^2}\right).$$

The log-likelihood for the full sample (conditioning on the x_i) is:

$$l(\beta, \alpha) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \sum_{i=1}^n \log \sigma_{x_i}^2 - \frac{1}{2} \sum_{i=1}^n \frac{(y_i - \beta_0 - \beta_1 x_i)^2}{\sigma_{x_i}^2}.$$

Both β and α appear: β through the residual $y_i - \beta_0 - \beta_1 x_i$ in the last sum, and α through $\sigma_{x_i}^2 = h(\alpha_0 + \alpha_1 x_i)$ in both the second and third terms. This is exactly the log-likelihood used in the BP paper (their formula before the Theorem).

Remark 1 (The measure of X is a nuisance that drops out). In the random-regressor framework, the joint density of (Y_i, X_i) factorises as $p(y_i, x_i) = p(y_i | x_i; \beta, \alpha) \cdot p_X(x_i)$. The log-likelihood for the full sample is therefore

$$\sum_i \log p(y_i | x_i; \beta, \alpha) + \sum_i \log p_X(x_i).$$

The second sum does not depend on (β, α) , so it is irrelevant for any inference about those parameters. The distribution of X is a nuisance — it is never specified, and the likelihood analysis proceeds entirely through the conditional piece $p(y|x; \beta, \alpha)$. This is why neither BP nor Koenker say anything about the distribution of X .

The mixed terminology Boris identifies

Boris observes that the model as written combines elements from two papers. Let us be precise.

From BP (1979): The parametrisation $\sigma_t^2 = h(z_t' \alpha)$ with h a twice-differentiable positive function, and the direct distributional assumption $u_t | x_t \sim \mathcal{N}(0, \sigma_t^2)$. BP have no ε in their paper: the Gaussian property sits on u itself, not on a standardised quantity. The log-likelihood above is BP's likelihood verbatim.

From Koenker (1981): The product structure $u_i = \sigma_i \varepsilon_i$ where ε_i are i.i.d. from an *unspecified* distribution F with mean zero, variance σ^2 , and finite fourth moment. F is **not** required to be Gaussian — this is Koenker's central generalisation. When F is Gaussian, Koenker's model coincides with BP's; when F is non-Gaussian, the BP test statistic has incorrect size (Koenker's main result).

The mixed model Boris writes uses the product form $U = \sigma_X \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 1)$ and $\sigma_X^2 = h(\alpha_0 + \alpha_1 X)$. This is *BP's model rewritten in product notation* — it is not Koenker's model. Koenker's model replaces $\varepsilon \sim \mathcal{N}(0, 1)$ with $\varepsilon \sim F$ (non-Gaussian).

Remark 2 (Correction to note 58). The Probabilist points out that in note 58 the Gaussian property was attributed incorrectly. The corrected picture: BP specify u as conditionally Gaussian directly, with no ε ; Koenker introduces ε precisely to drop the Gaussian assumption. Writing the mixed model with $\varepsilon \sim \mathcal{N}(0, 1)$ recovers BP, not Koenker.

Remark 3 (Summary of parameters and their roles).

Object	Random or deterministic	Observed?
X_i, Y_i	Random (in RR framework)	Yes
U_i, ε_i	Random	No
σ_{X_i}	Random (function of X_i)	No
β_0, β_1	Deterministic, unknown	No
α_0, α_1	Deterministic, unknown	No
$h(\cdot)$	Deterministic, known	—
Distribution of X	— (nuisance)	Irrelevant for l

The likelihood $l(\beta, \alpha)$ depends on both parameter vectors: β through the residuals, α through the scale function.

1.2 Likelihood of a pivot

We follow Boris’s notes of July 7, 2026 (note 59).

Warm-up: the location-normal family

Before tackling the BP model, consider the simplest possible case. Let $X \sim \mathcal{N}(\mu, 1)$ with $\mu \in \mathbb{R}$ unknown and variance fixed at 1. We observe a single value x .

The **pivot** is

$$Z = X - \mu.$$

Its distribution is $\mathcal{N}(0, 1)$ for every value of μ — fixed, no parameters. The density of Z is

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2},$$

a completely parameter-free function.

Now form the likelihood of μ given the observed x . The Jacobian of the map $x \mapsto z = x - \mu$ is $\partial z / \partial x = 1$ — a pure shift has no scale change. Therefore:

$$L(\mu \mid x) = \phi(x - \mu) \cdot 1 = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2}\right).$$

The log-likelihood is $l(\mu) = \text{const} - \frac{1}{2}(x - \mu)^2$, maximised at $\hat{\mu} = x$.

Let us make Boris’s observation precise in this simple case.

The distribution of Z does not depend on μ — true: $Z \sim \mathcal{N}(0, 1)$ always.

But Z still appears in the argument of the density — the observed value of the pivot is $z = x - \mu$, which depends on μ . When we evaluate the density at the observed value, we get $\phi(x - \mu)$, which is not constant.

One function does the work — in the location family the Jacobian is trivially 1, so there is only one source of parameter-dependence: the argument $x - \mu$ of ϕ . The “compensation” Boris describes is: as μ varies, the function $x \mapsto x - \mu$ shifts, and this shift is exactly what keeps the distribution of Z fixed. But for the likelihood we hold x fixed and vary μ — the shift no longer compensates, and the parameter-dependence in the argument is exposed.

Example 4. Take $x = 3$. Then:

μ	$z = x - \mu$	$\phi(z)$	$L(\mu)$
0	3.0	0.0044	0.0044
1	2.0	0.054	0.054
2	1.0	0.242	0.242
3	0.0	0.399	0.399
4	-1.0	0.242	0.242
5	-2.0	0.054	0.054

The likelihood peaks at $\hat{\mu} = x = 3$, where the pivot value is $z = 0$ (the mode of ϕ). At every row, $z = x - \mu$ is a different number — the distribution of Z is always $\mathcal{N}(0, 1)$, but which *value* we sit at in that distribution changes.

Remark 5 (For n i.i.d. observations). With $x_1, \dots, x_n$ i.i.d. $\mathcal{N}(\mu, 1)$, the pivots are $z_i = x_i - \mu$, all i.i.d. $\mathcal{N}(0, 1)$. The joint log-likelihood is

$$l(\mu) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2,$$

with MLE $\hat{\mu} = \bar{x}$. The score is

$$\frac{\partial l}{\partial \mu} = \sum_{i=1}^n (x_i - \mu),$$

which equals zero at $\hat{\mu} = \bar{x}$ and is centered ($E_\mu[\partial l / \partial \mu] = 0$) because $E_\mu[X_i] = \mu$.

Note 60: measure family, pivot map, and connecting L_X to L_Y

We follow Boris's notes of July 11, 2026 (note 60).

Setup. The location-normal **measure family** is $\{P_\mu : \mu \in \mathbb{R}\}$ where $P_\mu = \mathcal{N}(\mu, 1)$. The corresponding **density family** is $f_\mu(x) = \phi(x - \mu)$, where ϕ is the standard normal density. We observe a single realisation x of $X \sim P_\mu$.

The **pivot map** is

$$\varphi_\mu : x \mapsto y = x - \mu.$$

Applying it to the random variable: $Y := X - \mu$.

Proving the pivot has a fixed distribution. Using the CDF:

$$F_Y(y) = P_\mu(Y \leq y) = P_\mu(X - \mu \leq y) = P_\mu(X \leq y + \mu) = \Phi\left(\frac{(y + \mu) - \mu}{1}\right) = \Phi(y).$$

So $f_Y(y) = \phi(y)$ — the standard normal density, with no μ . This is the content of “pivot”: the law of Y is the same single measure $\mathcal{N}(0, 1)$ for every μ .

The likelihood question. Now observe $x_1, \dots, x_n$ i.i.d. from $\mathcal{N}(\mu, 1)$. Suppose we know μ . Then the pivot values are

$$y_i = x_i - \mu, \quad i = 1, \dots, n.$$

Consider the likelihood function for the y_i 's, viewed as observations from a parameter-free model $\mathcal{N}(0, 1)$:

$$L_Y(y_1, \dots, y_n) = \prod_{i=1}^n \phi(y_i) = (2\pi)^{-n/2} \exp\left(-\frac{1}{2} \sum_i y_i^2\right).$$

This is a fixed number once the y_i are fixed — it does not depend on μ , because the $\mathcal{N}(0, 1)$ model has no parameter. Boris notes: this is “a horizontal line” as a function of μ ... *but only if the y_i are held fixed.*

The likelihood of the original X -observations, viewed as a function of μ , is:

$$L_X(\mu) = \prod_{i=1}^n f_\mu(x_i) = \prod_{i=1}^n \phi(x_i - \mu).$$

How are L_X and L_Y connected? By substitution: $L_X(\mu) = L_Y(x_1 - \mu, \dots, x_n - \mu)$. The Jacobian of $x_i \mapsto y_i = x_i - \mu$ (a shift) is 1, so no extra factor appears.

The point Boris identifies: L_Y as a function of its arguments is flat in the sense of being a fixed formula $\prod \phi(\cdot)$. But when we substitute $y_i = x_i - \mu$ and think of the result as a function of μ , it is no longer flat — because the *arguments* $x_i - \mu$ slide as μ varies.

Special case $\bar{x} = \mu$. At the true $\mu = \bar{x}$ (assuming the sample mean equals μ for illustration), the pivot values are $y_i = x_i - \bar{x}$ — deviations from the mean, summing to zero. The likelihood evaluates to $\prod_i \phi(x_i - \bar{x})$, which is also the maximum of $L_X(\mu)$ over μ , since the MLE is $\hat{\mu} = \bar{x}$.

The verbalization Boris asks for: the fixed curve and the sliding point

Boris writes: “I only saw that the parameter disappears in the pivot. I knew it before. But I cannot illustrate what it means that the *value* of the random variable behind this pivot does depend on the parameter.”

Here is the illustration in words, then in a diagram.

Two different questions about $Y = X - \mu$:

1. *What is the distribution of Y ?*

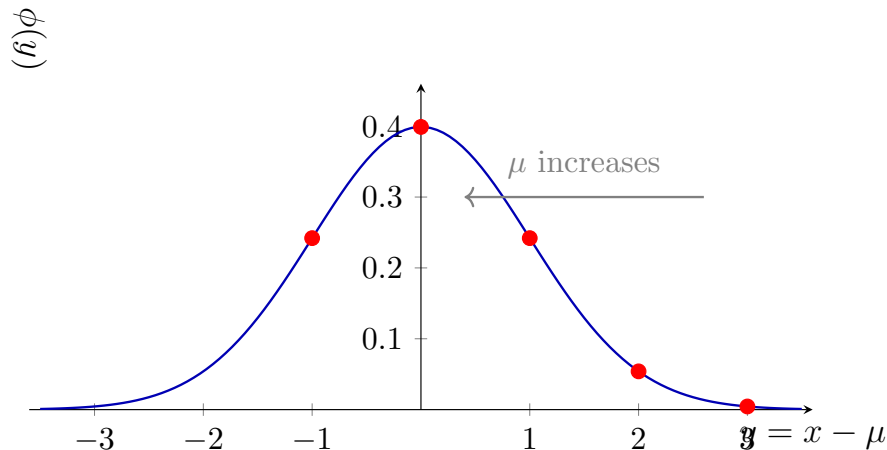
Answer: $\mathcal{N}(0, 1)$, for every μ . The distribution does not depend on μ .

2. *What value does Y take when we observe $X = x$?*

Answer: $Y = x - \mu$. This value *does* depend on μ .

These are different questions. Question 1 is about the law of the random variable (the density curve, which stays fixed). Question 2 is about where on that fixed curve a particular observation lands.

The sliding-point picture. Think of the standard normal density $\phi(y)$ as a fixed curve, drawn once and never changing. Now fix an observation $x = 3$. For each value of μ , the pivot maps this observation to the point $y = 3 - \mu$ on the horizontal axis. As μ increases, this point slides left along the fixed curve:



The **curve is always the same** $\mathcal{N}(0, 1)$ — the distribution of the pivot does not depend on μ . But the **red dot slides**: at $\mu = 0$ it sits far in the right tail (low density, low likelihood); at $\mu = 3 = x$ it sits at the peak (high density, maximum likelihood); at $\mu = 4$ it has slid to the left tail again (low density).

The likelihood $L_X(\mu) = \phi(x - \mu)$ is literally the height of the fixed curve at the sliding point. It depends on μ through the position of the dot, not through any change in the curve itself.

Remark 6 (Verbalization of what “value depends on parameter” means). “The distribution of Y does not depend on μ ” means: the *shape and position of the curve* are always the same standard normal, regardless of μ .

“The value of Y depends on μ ” means: fixing the observation ω (i.e., fixing $x = X(\omega)$), the number $Y(\omega) = x - \mu$ *changes* as μ changes. We slide along the fixed curve.

These two facts coexist without contradiction: the curve is fixed (as a function from $\mathbb{R}$ to $[0, \infty)$), but where on this curve we read off the likelihood is controlled by μ . The pivot construction ensures the curve is standard normal; the parameter μ controls the reading position.

The BP model: two functions instead of one

In the location family the Jacobian was 1: a shift does not rescale. The BP model adds a *scale* that depends on parameters, which brings in a non-trivial Jacobian. This is why Boris speaks of “two functions.”

The pivot is now

$$\varepsilon_i = \frac{y_i - \beta_0 - \beta_1 x_i}{\sigma_{x_i}}, \quad \sigma_{x_i}^2 = h(\alpha_0 + \alpha_1 x_i),$$

and $\varepsilon_i \sim \mathcal{N}(0, 1)$ for every (β, α) — still a pivot. The map $(y_i, x_i) \mapsto (\varepsilon_i, x_i)$ has Jacobian $\partial \varepsilon_i / \partial y_i = 1/\sigma_{x_i}$, now non-trivial. The likelihood contribution is

$$p(y_i | x_i; \beta, \alpha) = \phi(\varepsilon_i(\beta, \alpha)) \cdot \frac{1}{\sigma_{x_i}}.$$

The **two functions** Boris identifies:

1. $\phi(\varepsilon_i(\beta, \alpha))$ — the pivot density at the observed point; depends on (β, α) through the *argument*.
2. $1/\sigma_{x_i}$ — the Jacobian; depends on α through the *scale*.

In the location example, function (2) was identically 1 and dropped out. Here both contribute to the likelihood, and their product gives a richer dependence on (β, α) than either alone.

The “compensation” in the distribution of ε_i works as follows: as (β, α) change, the numerator $y - \beta_0 - \beta_1 x$ and the denominator σ_x both change — together they keep ε_i standard normal in distribution. In the likelihood, fixing (y_i, x_i) breaks this cooperation: the argument and the Jacobian evolve independently as (β, α) vary.

Suppose $\varepsilon_1, \dots, \varepsilon_n$ are i.i.d. $\mathcal{N}(0, 1)$. Their joint density on $\mathbb{R}^n$ is the product of n standard normal densities:

$$p_\varepsilon(\varepsilon_1, \dots, \varepsilon_n) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}} e^{-\varepsilon_i^2/2} = (2\pi)^{-n/2} \exp\left(-\frac{1}{2} \sum_{i=1}^n \varepsilon_i^2\right).$$

This is a fixed, parameter-free function on $\mathbb{R}^n$: it does not depend on β or α . Next time we will substitute $\varepsilon_i = (y_i - \beta_0 - \beta_1 x_i)/\sigma_{x_i}$ into this expression and account for the Jacobian to obtain the likelihood $l(\beta, \alpha)$.

The concept Boris identifies: likelihood of a pivot

Boris makes the following observation, which he says he never encountered explicitly in his studies.

A **pivot** is a function $T = t(X; \theta)$ of the data X and the parameter θ whose distribution does not depend on θ . The canonical example: if $X \sim \mathcal{N}(\mu, \sigma^2)$, then $Z = (X - \mu)/\sigma \sim \mathcal{N}(0, 1)$ for every (μ, σ) — the distribution of Z is fixed, but Z itself depends on the unknown parameters.

In our model, $\varepsilon_i = (y_i - \beta_0 - \beta_1 x_i)/\sigma_{x_i}$ is exactly a pivot: for every (β, α) ,

$$\varepsilon_i(\beta, \alpha) \sim \mathcal{N}(0, 1).$$

The distribution does not depend on parameters. Yet ε_i itself depends on them through the formula.

Now consider the likelihood. We evaluate the density of Y_i given $X_i = x_i$ using the change-of-variables formula:

$$p(y_i \mid x_i; \beta, \alpha) = \underbrace{\frac{1}{\sqrt{2\pi}} e^{-\varepsilon_i(\beta, \alpha)^2/2}}_{\text{pivot density at observed point}} \cdot \underbrace{\frac{1}{\sigma_{x_i}}}_{\text{Jacobian}}.$$

Boris's observation: there are two separate dependencies on the parameters.

- The density function $\phi(\cdot)$ is fixed (does not depend on parameters). But its *argument* $\varepsilon_i(\beta, \alpha)$ does.
- The Jacobian $1/\sigma_{x_i}$ depends on α through $\sigma_{x_i}^2 = h(\alpha_0 + \alpha_1 x_i)$.

Together, these two functions give a non-trivial, parameter-dependent likelihood — even though the distribution of the pivot is fixed.

Remark 7 (Why the two dependencies do not cancel in the likelihood). One might wonder: if the distribution of $\varepsilon_i(\beta, \alpha)$ is always $\mathcal{N}(0, 1)$, why does the likelihood depend on β and α at all?

The answer is that the distribution being fixed means: if you draw a new $\varepsilon_i(\beta, \alpha)$ for each (β, α) , you always get a $\mathcal{N}(0, 1)$ value. But in the likelihood we do not draw a new ε_i — we fix the *observed data* (y_i, x_i) and vary (β, α) . Different parameter values assign different values to $\varepsilon_i(y_i, x_i; \beta, \alpha)$ — some close to zero (high density), some far (low density). The likelihood measures how plausible the observed data are under each parameter value; the pivot construction is a tool for doing this calculation cleanly.

Concretely: at the true (β_0, α_0) , the observed y_i was generated with $\varepsilon_i \approx 0$ (on average), so the density is high. At wrong parameter values, the implied ε_i may be large, and the density is low.

Remark 8 (Boris's formulation: two functions compensating). Boris writes: “distribution does not depend on parameters as a function, but it still appears in its arguments. Two functions compensate each other giving the fixed density for the whole family.”

The two functions are: (i) the map $(\beta, \alpha) \mapsto \varepsilon_i(\beta, \alpha)$ (shifts the argument of ϕ), and (ii) the Jacobian $1/\sigma_{x_i}$ (scales the density). They compensate in the following precise sense: for any fixed (β, α) , the pushforward of the law of Y_i (given $X_i = x_i$) under the map $y \mapsto (y - \beta_0 - \beta_1 x_i)/\sigma_{x_i}$ is $\mathcal{N}(0, 1)$. This is the content of the pivot: the two functions cooperate to produce a parameter-free distribution. But in the likelihood, we evaluate at a fixed y_i , not at a draw from the distribution, so the compensation is broken and the parameter-dependence is exposed.

Plugging in: from joint pivot density to the likelihood

Boris's homework set up the joint density of $\varepsilon_1, \dots, \varepsilon_n$. The next step: substitute

$$\varepsilon_i = \varepsilon_i(\beta, \alpha) := \frac{y_i - \beta_0 - \beta_1 x_i}{\sigma_{x_i}}, \quad \sigma_{x_i}^2 = h(\alpha_0 + \alpha_1 x_i),$$

and account for the Jacobian of the vector map $(y_1, \dots, y_n) \mapsto (\varepsilon_1, \dots, \varepsilon_n)$:

$$\det\left(\frac{\partial \varepsilon}{\partial y}\right) = \prod_{i=1}^n \frac{1}{\sigma_{x_i}}.$$

(The map is diagonal, so the Jacobian determinant is the product of the individual Jacobians.) The log-likelihood is:

$$\begin{aligned} l(\beta, \alpha) &= \log p_\varepsilon(\varepsilon_1(\beta, \alpha), \dots, \varepsilon_n(\beta, \alpha)) + \log \prod_{i=1}^n \frac{1}{\sigma_{x_i}} \\ &= -\frac{n}{2} \log(2\pi) - \frac{1}{2} \sum_{i=1}^n \varepsilon_i(\beta, \alpha)^2 - \sum_{i=1}^n \log \sigma_{x_i} \\ &= -\frac{n}{2} \log(2\pi) - \frac{1}{2} \sum_{i=1}^n \log \sigma_{x_i}^2 - \frac{1}{2} \sum_{i=1}^n \frac{(y_i - \beta_0 - \beta_1 x_i)^2}{\sigma_{x_i}^2}, \end{aligned}$$

which is the BP log-likelihood derived in § 1.1 by a different route. The two routes — direct density of $Y|X$ and change of variables through the pivot ε — give identical results, as they must.

1.3 The general pivot framework and its application to regression

We follow Boris's notes of July 12, 2026 (note 61).

Boris's framework: compensating map

Setup. Let $(\Omega, \mathcal{F})$ be a measurable space. We have a random variable

$$X : \Omega \longrightarrow \mathbb{R},$$

which is a *fixed* measurable map — it does not depend on any parameter. We have a parametrised family of measures on $\mathbb{R}$:

$$\{P_\theta\}_{\theta \in \Theta}.$$

This family tells us how X is distributed: $X \sim P_\theta$ when the true parameter is θ . The variability across θ lives entirely in the measures P_θ , not in the map X .

Suppose densities exist. Then $\{P_\theta\}$ is connected to a parametrised family of densities $\{f_\theta\}$ via a base (reference) measure on $\mathbb{R}$. Two standard cases:

- **Continuous:** $f_\theta(x)$ is a density with respect to Lebesgue measure on $\mathbb{R}$, so $P_\theta(A) = \int_A f_\theta(x) dx$.
- **Discrete:** $f_\theta(x)$ is a mass function with respect to counting measure on some countable set, so $P_\theta(A) = \sum_{x \in A} f_\theta(x)$.

On Ω we introduce *no* such base measure and therefore *no* density on Ω . We do not have an analogue of Lebesgue measure sitting on Ω , so the map $X : \Omega \rightarrow \mathbb{R}$ is not a density — it is simply a measurable function. Densities live on $\mathbb{R}$ (the x -axis), not on Ω .

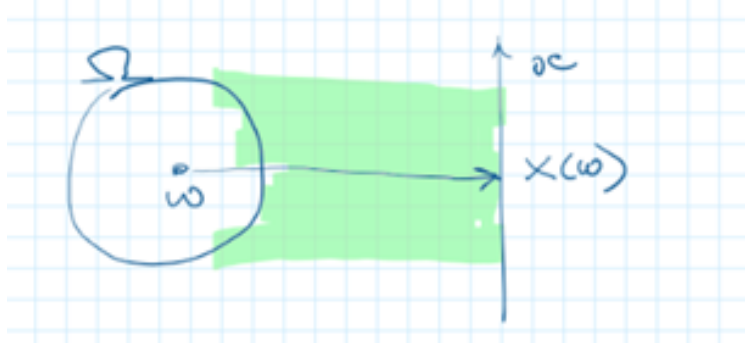


Figure 1: Boris's diagram (July 12, 2026): the fixed map $X : \Omega \rightarrow \mathbb{R}$ sending ω to $X(\omega)$. The oval is Ω ; the green band is the image of X .

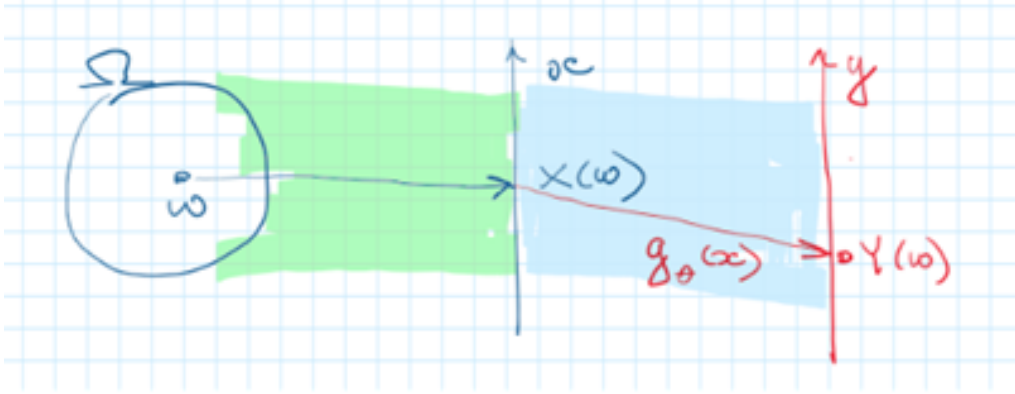


Figure 2: Boris's diagram (July 12, 2026): the full composition $\Omega \xrightarrow{X} \mathbb{R} \xrightarrow{g_\theta} \mathbb{R}$, sending $\omega \mapsto X(\omega) \mapsto Y(\omega) = g_\theta(X(\omega))$. The second axis (red) carries $Y(\omega)$.

The compensating map. We introduce a parametrised family of measurable maps

$$g_\theta : \mathbb{R} \longrightarrow \mathbb{R}, \quad \theta \in \Theta.$$

Equivalently, g is a function of two variables: $g : \mathbb{R} \times \Theta \rightarrow \mathbb{R}$. Define the composed random variable

$$Y_\theta(\omega) := g_\theta(X(\omega)).$$

This is a new random variable, parametrised by θ through the map g_θ .

Its distribution is the pushforward

$$Q_\theta := (g_\theta)_* P_\theta,$$

meaning $Q_\theta(B) = P_\theta(\{x : g_\theta(x) \in B\})$ for any Borel set B .

Definition of a pivot. If Q_θ does not depend on θ — that is, the image of the whole measure family $\{P_\theta\}$ under $\{g_\theta\}$ is a single measure Q — then Y_θ is called a **pivot variable** (or **pivotal quantity**) with respect to the family $\{P_\theta\}$:

$$(g_\theta)_* P_\theta = Q \quad \text{for all } \theta \in \Theta.$$

The family $\{g_\theta\}$ is the *compensating map*: it corrects for the variation in P_θ so that the resulting distribution is always Q .

Why pivot values must depend on θ . Boris raises the point from note 60: while the *distribution* of Y_θ is parameter-free, its *value* $Y_\theta(\omega) = g_\theta(X(\omega))$ generically depends on θ . The abstract framework makes this a necessity, not an accident.

Suppose g_θ is non-trivial: there exist $\theta \neq \theta'$ and a set A with $P_\theta(A) > 0$ such that $g_\theta \neq g_{\theta'}$ on A . Since $(g_\theta)_*P_\theta = (g_{\theta'})_*P_{\theta'} = Q$ but $P_\theta \neq P_{\theta'}$ (the family is non-trivial), the map g_θ must absorb the difference between P_θ and $P_{\theta'}$. That absorption is precisely the variation in g_θ . For a fixed ω , the value $g_\theta(X(\omega))$ must therefore vary with θ — otherwise the compensation is impossible.

Remark 9 (Boris’s range example). Boris proposes abstracting from the measure to a simpler operator, for instance the range $\text{rng}(P) = \max - \min$ of the support. If P_θ has range $r(\theta)$, one could choose $g_\theta(x) = x/r(\theta)$ to stabilise the range of Q_θ . The map still depends on θ — the compensating dependence is unavoidable. Similarly, for symmetric distributions $p \sim \mathcal{N}(0, \sigma^2)$, the map $x \mapsto -x$ preserves the distribution but changes values.

The Probabilist’s framework: inverse map

The Probabilist proposes a dual viewpoint that starts from the other end.

Setup. Fix a single measure P on Ω and a single fixed map

$$X : \Omega \longrightarrow \mathbb{R}.$$

The distribution of X is fixed: $P_X := X_*P$ is a single measure on $\mathbb{R}$ that does not depend on any parameter. Think of X as the “primordial” randomness — a standard source of noise.

The generating map. Now introduce the same family $\{g_\theta : \mathbb{R} \rightarrow \mathbb{R}\}_{\theta \in \Theta}$. Define the composed variable

$$Y_\theta(\omega) := g_\theta(X(\omega)).$$

In the Probabilist’s framework, the distribution of Y_θ *does* depend on θ , because g_θ introduces the parameter:

$$Q_\theta := (g_\theta)_*P_X \quad \text{varies with } \theta.$$

The family $\{Q_\theta\}$ is the observed distribution family. We observe Y (the data), and X is latent.

Recovery: the pivot. If g_θ is invertible (for each θ), we can recover X from Y_θ by the inverse map:

$$X(\omega) = g_\theta^{-1}(Y_\theta(\omega)).$$

Substituting the observed $Y = y$:

$$x = g_\theta^{-1}(y).$$

The quantity $g_\theta^{-1}(Y)$ — which depends on both the observation Y and the parameter θ — has a fixed distribution P_X regardless of θ . So $g_\theta^{-1}(Y)$ is the pivot: it is X itself, reconstructed from the data.

Remark 10 (Comparing the two frameworks). The two frameworks are dual and connected by the invertibility of g_θ :

	Boris’s framework	the Probabilist’s framework
Primary object	$\{P_\theta\}$ on $\mathbb{R}$	Fixed P on Ω
Fixed quantity	Map X	Map X , measure P
g_θ role	Compensating (reduces to Q)	Generating (produces $\{Q_\theta\}$)
Pivot	$Y_\theta = g_\theta(X)$, dist. Q	$X = g_\theta^{-1}(Y)$, dist. P_X
What we observe	$X \sim P_\theta$	$Y \sim Q_\theta$

In Boris’s framework the pivot lives on the *output* of g_θ ; in the Probabilist’s framework the pivot lives on the *input* of g_θ , recovered by inversion. They are two descriptions of the same mathematical object.

Application to basic regression

We apply both frameworks to the simplest regression model: one-dimensional, no heteroscedasticity.

The model. Observe $(X_1, Y_1), \dots, (X_n, Y_n)$ where

$$Y_i = \beta_0 + \beta_1 X_i + \sigma \varepsilon_i,$$

with X_i fixed design points and $\varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$. The parameter is $\theta = (\beta_0, \beta_1, \sigma)$.

The Probabilist’s framework fits naturally. Set $\Omega = \mathbb{R}^n$, the fixed measure $P = \mathcal{N}(0, 1)^{\otimes n}$, and the fixed random vector

$$\varepsilon = (\varepsilon_1, \dots, \varepsilon_n) : \Omega \longrightarrow \mathbb{R}^n.$$

The distribution of ε under P is $\mathcal{N}(0, I_n)$ — fixed, no parameters. This is the “primordial” source of randomness.

The generating map $g_\theta : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is

$$g_\theta(\varepsilon_1, \dots, \varepsilon_n) = (\beta_0 + \beta_1 X_1 + \sigma \varepsilon_1, \dots, \beta_0 + \beta_1 X_n + \sigma \varepsilon_n) =: Y.$$

The observation $Y = g_\theta(\varepsilon)$ has distribution $\mathcal{N}(\beta_0 \mathbf{1} + \beta_1 \mathbf{x}, \sigma^2 I_n)$ — depending on θ . The pivot is recovered by inversion:

$$\varepsilon_i = g_\theta^{-1}(Y)_i = \frac{Y_i - \beta_0 - \beta_1 X_i}{\sigma}.$$

This quantity has distribution $\mathcal{N}(0, 1)$ for every θ — it is the pivot. In regression language: *the standardised errors are the pivot*.

Boris’s framework for the same model. The observed quantity is Y with distribution family $\{P_\theta\}_{\theta \in \Theta}$ where $P_\theta = \mathcal{N}(\beta_0 \mathbf{1} + \beta_1 \mathbf{x}, \sigma^2 I_n)$.

The compensating map:

$$g_\theta(y_1, \dots, y_n) = \left(\frac{y_1 - \beta_0 - \beta_1 X_1}{\sigma}, \dots, \frac{y_n - \beta_0 - \beta_1 X_n}{\sigma} \right).$$

The pushforward $(g_\theta)_* P_\theta = \mathcal{N}(0, I_n)$ for all θ — a single fixed measure. So $g_\theta(Y)$ is the pivot.

Both frameworks give the same pivot $\varepsilon_i = (Y_i - \beta_0 - \beta_1 X_i)/\sigma$, differing only in which direction the map g_θ is applied. The Probabilist’s framework starts from ε and generates Y ; Boris’s framework starts from Y and maps back to ε .

Why the pivot values depend on θ . For a fixed draw ω (i.e., a fixed $\varepsilon_i = c$), the observation is $Y_i(\omega) = \beta_0 + \beta_1 X_i + \sigma c$, which depends on θ . Conversely, for a fixed observation $Y_i = y$, the pivot value is $\varepsilon_i = (y - \beta_0 - \beta_1 X_i)/\sigma$, which also depends on θ .

The compensating structure: when σ increases, P_θ spreads out (larger variance), but g_θ divides by the larger σ , squeezing the distribution back to $\mathcal{N}(0, 1)$. When β_0 shifts, P_θ shifts, but g_θ subtracts the shift. Each change in θ deforms P_θ , and g_θ counteracts this deformation. The values $g_\theta(y)$ must change with θ for this counteraction to work — exactly Boris’s point from note 61.

Remark 11 (The BP pivot as the next step). In basic regression, both the location (β) and scale (σ) are constant across observations: the same $(\beta_0, \beta_1, \sigma)$ applies to every i . The compensating map g_θ is the same function applied to each (Y_i, X_i) pair.

In the BP model, the scale $\sigma_i = \sqrt{h(\alpha_0 + \alpha_1 X_i)}$ varies with X_i . The compensating map becomes

$$g_\theta(Y_i, X_i) = \frac{Y_i - \beta_0 - \beta_1 X_i}{\sqrt{h(\alpha_0 + \alpha_1 X_i)}},$$

where now g_θ depends on *two* parameter blocks: β (through the location) and α (through the scale function h). The pivot $\varepsilon_i = g_\theta(Y_i, X_i)$ is still $\mathcal{N}(0, 1)$ for all (β, α) — but recovering it from the data requires knowing (or estimating) both parameter blocks.

1.4 Pivot in regression: the score argument

We follow Boris's notes of July 13, 2026 (note 62).

Basic regression: did I construct a pivot?

Boris considers the single-observation model:

$$Y = \beta_0 + \beta_1 X + U, \quad U = \sigma \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1),$$

where $\sigma > 0$ is a deterministic constant (not random). Expressing ε through the observables:

$$\varepsilon = \frac{Y - \beta_0 - \beta_1 X}{\sigma}.$$

Boris asks: did I construct a pivot?

The Probabilist: Yes — but it is a tautology. The model *defines* $\varepsilon = U/\sigma$, so the standardised error is parameter-free by construction. There is nothing to prove: the pivot property follows immediately from the definition.

Nevertheless, the construction is correct. For n observations indexed by i , the pivot values are

$$\varepsilon_i = \frac{Y_i - \beta_0 - \beta_1 X_i}{\sigma}, \quad i = 1, \dots, n,$$

each $\mathcal{N}(0, 1)$ independently of the parameter $(\beta_0, \beta_1, \sigma)$.

The BP pivot

Boris's hypothesis: the BP pivot for observation i is

$$\varepsilon_i = \frac{Y_i - \beta_0 - \beta_1 X_i}{\sqrt{h(\alpha^\top z_i)}}.$$

The Probabilist: Yes, this is correct — provided $h(\cdot)$ is parametrised by α . Each ε_i is $\mathcal{N}(0, 1)$ for every (β, α) .

However, the Probabilist notes a conceptual jump: in basic regression, σ is a constant (deterministic, the same for all i). In the BP model, $\sigma_i = \sqrt{h(\alpha^\top z_i)}$ varies with i because z_i varies — and if $z_i = x_i$ is a realisation of a random X_i , then σ_i is itself random. This is the transition from deterministic to random scale discussed in notes 50–51.

Why the score is not flat zero: the key argument

Boris: if $\varepsilon \sim \mathcal{N}(0, 1)$ and its density does not depend on parameters, then its score is identically zero. So where does the non-trivial score come from?

The Probabilist: If we *observed* ε_i directly, then yes — the likelihood would be

$$L(\varepsilon_1, \dots, \varepsilon_n) = \prod_{i=1}^n \phi(\varepsilon_i),$$

which contains no parameters. The score with respect to any θ would be identically zero. But we do not observe ε_i . We observe Y_i and X_i .

The resolution: ε_i is not a free variable — it is a *function* of the observed data and the unknown parameters: $\varepsilon_i = \varepsilon_i(Y_i, X_i; \theta)$. When we substitute this expression into the likelihood, the parameters reappear through the argument of ϕ .

Abstract formulation: composition of two functions

The Probabilist states the argument abstractly. Let f be a function (the density ϕ) and let g_θ be the pivot map (depending on θ). Consider the composition

$$h(\theta) := f(g_\theta(y)).$$

This composition depends on θ — so its derivative with respect to θ is generally non-zero.

Set $\eta := g_\theta(y)$ (the pivot value at the observation y). There are two ways to look at η :

View 1 — η as an independent variable. Treat η as a free number. Then

$$\frac{\partial f}{\partial \eta}(\eta) = \phi'(\eta) = -\eta \phi(\eta).$$

This is a property of f alone; no θ involved.

View 2 — η as a dependent variable ($\eta = g_\theta(y)$, function of θ). By the chain rule:

$$\frac{\partial}{\partial \theta} f(g_\theta(y)) = f'(g_\theta(y)) \cdot \frac{\partial g_\theta(y)}{\partial \theta} = \phi'(\varepsilon(\theta)) \cdot \frac{\partial \varepsilon}{\partial \theta}.$$

This is generally non-zero because $\partial \varepsilon / \partial \theta \neq 0$: the pivot value changes when θ changes (it must, to keep the distribution fixed).

In regression, $\varepsilon_i(\theta) = (y_i - \beta_0 - \beta_1 x_i) / \sigma$ gives

$$\frac{\partial \varepsilon_i}{\partial \beta_0} = -\frac{1}{\sigma}, \quad \frac{\partial \varepsilon_i}{\partial \beta_1} = -\frac{x_i}{\sigma}, \quad \frac{\partial \varepsilon_i}{\partial \sigma} = -\frac{y_i - \beta_0 - \beta_1 x_i}{\sigma^2} = -\frac{\varepsilon_i}{\sigma}.$$

None of these are zero in general.

The log-likelihood through substitution

Substituting $\varepsilon_i = (y_i - \beta_0 - \beta_1 x_i) / \sigma$ into the joint density of $(\varepsilon_1, \dots, \varepsilon_n)$ and including the Jacobian $\prod_i (1/\sigma)$, the likelihood is:

$$L(\beta_0, \beta_1, \sigma; y_1, \dots, y_n) = \prod_{i=1}^n \phi\left(\frac{y_i - \beta_0 - \beta_1 x_i}{\sigma}\right) \cdot \frac{1}{\sigma}.$$

Taking logarithms, the log-likelihood is

$$\begin{aligned}
l(\beta_0, \beta_1, \sigma) &= \sum_{i=1}^n \log \phi(\varepsilon_i) - n \log \sigma \\
&= -\frac{n}{2} \log(2\pi) - \frac{1}{2} \sum_{i=1}^n \varepsilon_i^2 - n \log \sigma \\
&= -\frac{n}{2} \log(2\pi) - n \log \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2.
\end{aligned}$$

This is the standard normal regression log-likelihood, and it manifestly depends on $(\beta_0, \beta_1, \sigma)$ through both the residuals and the $-n \log \sigma$ term.

Remark 12 (The two contributions to the score). Differentiating l with respect to β_0 :

$$\frac{\partial l}{\partial \beta_0} = \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = \frac{1}{\sigma} \sum_{i=1}^n \varepsilon_i.$$

This is the chain rule in action: $\partial l / \partial \beta_0 = \sum_i (\partial \log \phi / \partial \varepsilon_i) \cdot (\partial \varepsilon_i / \partial \beta_0) = \sum_i (-\varepsilon_i) \cdot (-1/\sigma)$.

The score is zero at the MLE $\hat{\beta}_0$ because $\sum_i \hat{\varepsilon}_i = 0$ (OLS residuals sum to zero). This is exactly the condition that the true ε_i are centered — it holds at the MLE, not in general.

1.5 Transformed likelihood and non-injectivity of the pivot map

We follow Boris's notes of July 14, 2026 (note 63).

Likelihood as probability of the observed event

For a cleaner conceptual foundation, start in the discrete case. Let $X_1, \dots, X_n$ be i.i.d. with probability mass function $p(x; \theta) = P_\theta\{X = x\}$. The likelihood is

$$L(\theta; x_1, \dots, x_n) = p(x_1; \theta) \cdots p(x_n; \theta) = P_\theta\{X_1 = x_1, \dots, X_n = x_n\}. \quad (1)$$

This has a direct interpretation: it is the probability, under measure P_θ , of observing the exact values we did. The MLE is the θ that makes this probability largest — it selects the measure under which the observed data are most plausible.

Transformed likelihood: the bijective case

Suppose there is a transformation g such that $g(X)$ has a more standard or tractable distribution (e.g. a pivot). Consider the alternative expression

$$P_\theta\{g(X_1) = g(x_1), \dots, g(X_n) = g(x_n)\}. \quad (2)$$

If g is bijective: the events in (1) and (2) are identical — $g(X_i) = g(x_i) \Leftrightarrow X_i = x_i$ — so (1) and (2) are the same number, and maximising either over θ gives the same MLE. The benefit is purely presentational: the distribution of $g(X)$ may be tabulated or familiar, making it easier to work with.

Regression: the map is not injective

In the regression setting, the observed unit is the pair $(X_i, Y_i) \in \mathbb{R}^2$. Its joint distribution depends on (β_0, β_1) and on the unknown marginal distribution of X_i — which we do not specify. Writing the joint likelihood of (x_i, y_i) pairs directly is therefore impossible.

The natural transformation is

$$g_\beta(x, y) = y - \beta_0 - \beta_1 x = \varepsilon,$$

a map $g_\beta : \mathbb{R}^2 \rightarrow \mathbb{R}$. Under this map, the residual $\varepsilon_i = Y_i - \beta_0 - \beta_1 X_i$ has distribution $\mathcal{N}(0, \sigma^2)$ — one parameter instead of the original three $(\beta_0, \beta_1, \sigma)$. The parameters β_0 and β_1 have been “absorbed” into the transformation; only σ remains visible.

Moreover, $\varepsilon_1, \dots, \varepsilon_n$ are i.i.d. (because (X_i, Y_i) are i.i.d. and the transformation is the same for each i), so the transformed likelihood is again a product:

$$L(\sigma; \varepsilon_1(\beta), \dots, \varepsilon_n(\beta)) = \prod_{i=1}^n f_\sigma(\varepsilon_i(\beta)).$$

The problem: g_β is not injective

Here a conceptual difficulty arises. The map $g_\beta : \mathbb{R}^2 \rightarrow \mathbb{R}$ is affine and *reduces dimension*, so it cannot be injective:

- If $\beta_1 = 0$: any two pairs (y, x_1) and (y, x_2) with the same y map to the same value $y - \beta_0$. Non-injectivity is obvious.
- If $\beta_1 \neq 0$: the kernel of the linear part of g_β is the line $\{(x, y) : y = \beta_1 x\}$ in $\mathbb{R}^2$. Every fibre $g_\beta^{-1}(\varepsilon)$ is the line $\{(x, y) : y - \beta_0 - \beta_1 x = \varepsilon\}$ — an infinite set. Injectivity fails.

Since any linear map from $\mathbb{R}^2$ to $\mathbb{R}$ has a non-trivial kernel, injectivity is structurally impossible.

Consequence for the likelihood. The equivalence between (1) and (2) required bijectivity. Without it, the event $\{g_\beta(X_i, Y_i) = \varepsilon_i\}$ contains infinitely many pairs, while the event $\{(X_i, Y_i) = (x_i, y_i)\}$ is a single point. So

$$P_\theta\{g_\beta(X_i, Y_i) = \varepsilon_i\} \neq P_\theta\{(X_i, Y_i) = (x_i, y_i)\},$$

and the MLE from the transformed likelihood is generally *different* from the MLE of the original.

Resolution: it is the only likelihood we can write

Boris identifies the resolution. Although the transformed likelihood is not equivalent to the joint likelihood of (X_i, Y_i) pairs, it is *the only likelihood available in this model*. The reason: we place no restrictions on the marginal distribution of X_i . The distribution of X is a completely free infinite-dimensional nuisance. Without specifying it, the joint distribution of (X_i, Y_i) cannot be written down, and neither can the full joint likelihood (1).

What *can* be written is the conditional likelihood:

$$L_{\text{cond}}(\beta, \sigma; y_1, \dots, y_n \mid x_1, \dots, x_n) = \prod_{i=1}^n f_{Y|X}(y_i \mid x_i; \beta, \sigma) = \prod_{i=1}^n \frac{1}{\sigma} \phi\left(\frac{y_i - \beta_0 - \beta_1 x_i}{\sigma}\right).$$

This is exactly the product of the transformed densities (with Jacobian $1/\sigma$ from dividing by σ). Maximising this over (β, σ) gives OLS estimates of β and an estimate of σ^2 — the standard regression procedure.

Remark 13 (Why the non-injectivity does not invalidate the method). The non-injectivity of g_β means the transformed likelihood is not the same number as the joint likelihood. But we are not claiming equivalence to a joint likelihood that we cannot write anyway. We are simply maximising the conditional likelihood, which is the natural object given that the marginal distribution of X is unspecified. The OLS estimator $\hat{\beta}$ is the MLE of this conditional likelihood, and it has the usual optimal properties under the model assumptions.

Remark 14 (Connection to the pivot framework). In the language of § 1.3, the map $g_\beta(x, y) = y - \beta_0 - \beta_1 x$ is Boris's compensating map restricted to the y -direction. It sends the family of conditional distributions $\{f_{Y|X=x}(\cdot; \beta)\}$ to the fixed distribution $\mathcal{N}(0, \sigma^2)$. The Jacobian of the inverse (from ε back to y , holding x fixed) is σ — this is where the $1/\sigma$ factor in the conditional density comes from.

1.6 Likelihood as a surface and random curtains

We follow Boris's notes of July 15, 2026 (note 64).

The likelihood surface

For the location-normal family with a single observation, both the parameter μ and the observation x are real numbers. The likelihood can be viewed as a function of *both* simultaneously:

$$L(x, \mu) = \phi(x - \mu) = \frac{1}{\sqrt{2\pi}} e^{-(x-\mu)^2/2}.$$

Boris notes this is not the standard view — usually one fixes the observation and varies μ , or fixes μ and varies x — but looking at the full surface is valid and illuminating.

The surface $L(x, \mu)$ lives over the (x, μ) -plane. It has the shape of a ridge: a bell-shaped hill running along the diagonal $x = \mu$, falling off on both sides. Two natural cross-sections:

- **Fix** $\mu = \mu_0$, **vary** x : the section $x \mapsto \phi(x - \mu_0)$ is the density of $\mathcal{N}(\mu_0, 1)$ — a bell centred at $x = \mu_0$.
- **Fix** $x = x_0$, **vary** μ : the section $\mu \mapsto \phi(x_0 - \mu)$ is the likelihood at observation x_0 — a bell centred at $\mu = x_0$.

Boris observes: both sections have the *same shape*. The Probabilist's explanation: for any symmetric location family with density $f_0(x - \mu)$, the two sections are

$$f_0(x - \mu_0) \quad (\text{density section}) \quad \text{and} \quad f_0(x_0 - \mu) = f_0(-(\mu - x_0)) = f_0(\mu - x_0) \quad (\text{likelihood section}).$$

When f_0 is symmetric ($f_0(-z) = f_0(z)$), both equal f_0 of the absolute difference $|x - \mu|$, giving identical shapes. This holds for the normal and the Cauchy, but not for asymmetric families (exponential, Poisson): for those, the density and likelihood cross-sections have different shapes.

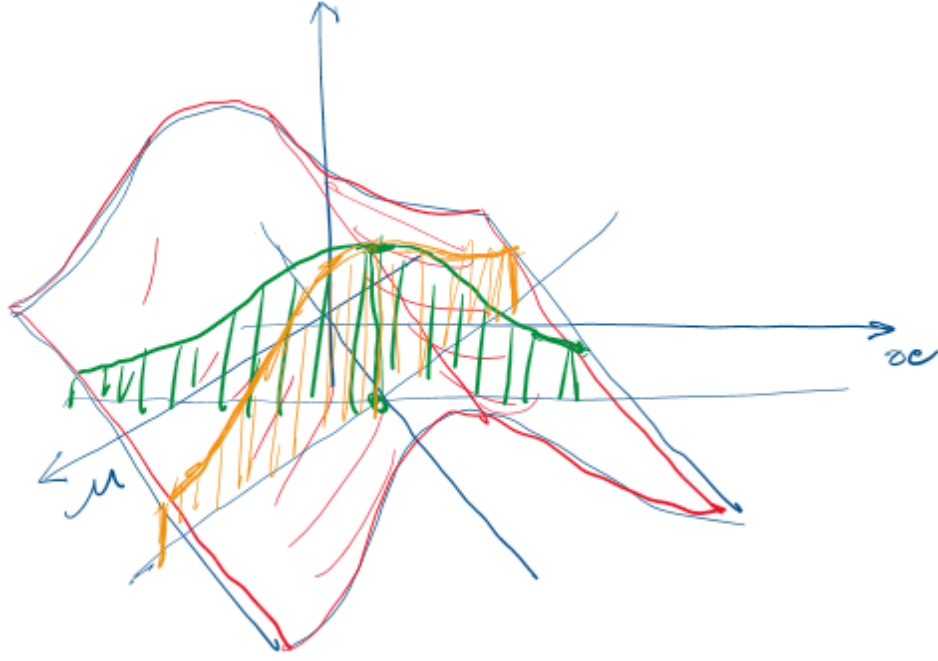


Figure 3: Boris’s diagram 1 (July 15, 2026): the 3D surface $L(x, \mu) = \phi(x - \mu)$ over the (x, μ) -plane. The ridge runs along the diagonal $x = \mu$. Sections by fixed x give densities; sections by fixed μ give likelihoods — both are bell-shaped with the same width.

Orange curtains: random likelihood curves

Fix the true parameter μ_t . Sample observations $x_1, x_2, \dots$ from $\mathcal{N}(\mu_t, 1)$. Each observation x_i is a point on the x -axis; Boris draws them as orange dots, dense near μ_t and sparse far from it (reflecting the bell-shaped sampling density).

Each observation x_i generates a likelihood curve: $\mu \mapsto \phi(x_i - \mu)$ — a bell centred at $\mu = x_i$. Boris calls these the **orange curtains**: vertical slices of the surface $L(x, \mu)$ at $x = x_i$, “hanging” from the ridge. Since observations x_i cluster near μ_t , the orange curtains also cluster near μ_t , and their peaks concentrate around $\mu = \mu_t$.

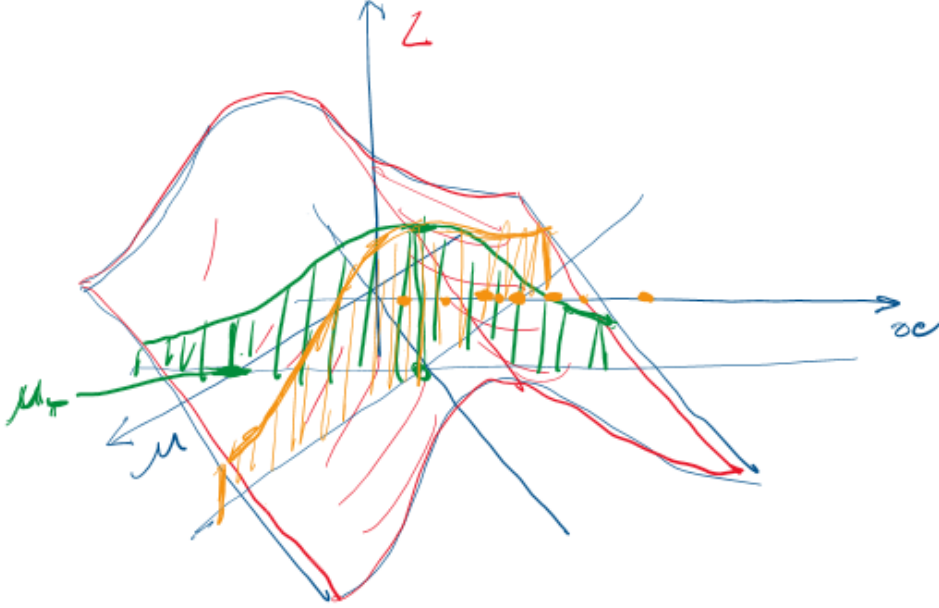


Figure 4: Boris’s diagram 2: the same surface $\phi(x - \mu)$ with orange sample points drawn from $\mathcal{N}(\mu_t, 1)$ on the x -axis. Each point gives rise to an orange likelihood curtain. Curtains cluster near μ_t .

Remark 15 (Naming “true” vs. “abstract” μ). Boris notes the need to distinguish two roles of μ . The *true* parameter μ_t is fixed and governs the sampling of x . The *abstract* (or hypothetical, or dummy) μ is the variable over which we draw the likelihood curve. In frequentist statistics there is no standard single name for the dummy μ ; it is often called the “parameter argument” or simply “the parameter” with the understanding that the likelihood is a function of it. The true μ_t is a fixed unknown constant, not a variable.

The pivot surface

Now apply the pivot map $g_\mu(x) = x - \mu$. Boris stresses: here μ is the abstract parameter, not necessarily the true one. Viewed as a function of two variables (x, μ) , the map

$$g : (x, \mu) \mapsto y = x - \mu$$

is a *plane* — it is linear (affine) in both x and μ simultaneously. This plane tilts at 45 across the (x, μ) -plane: where $x > \mu$ the output is positive; where $x < \mu$ it is negative; the zero-set is the diagonal $x = \mu$.

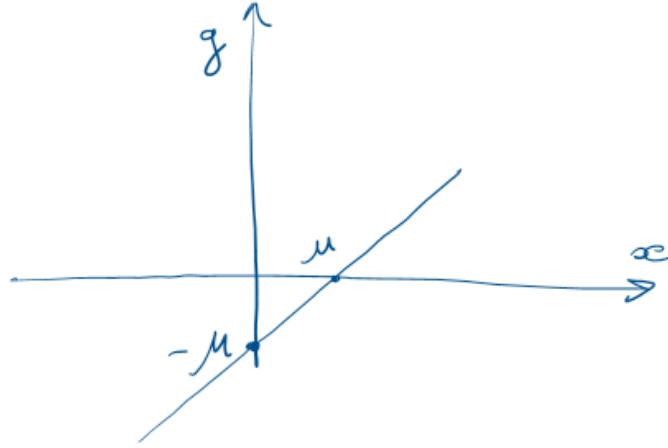


Figure 5: Boris's diagram 3: the 2D map $y = x - \mu$ as a function of (x, μ) . The shading shows positive (blue) and negative (red) regions separated by the diagonal $x = \mu$ where $y = 0$.

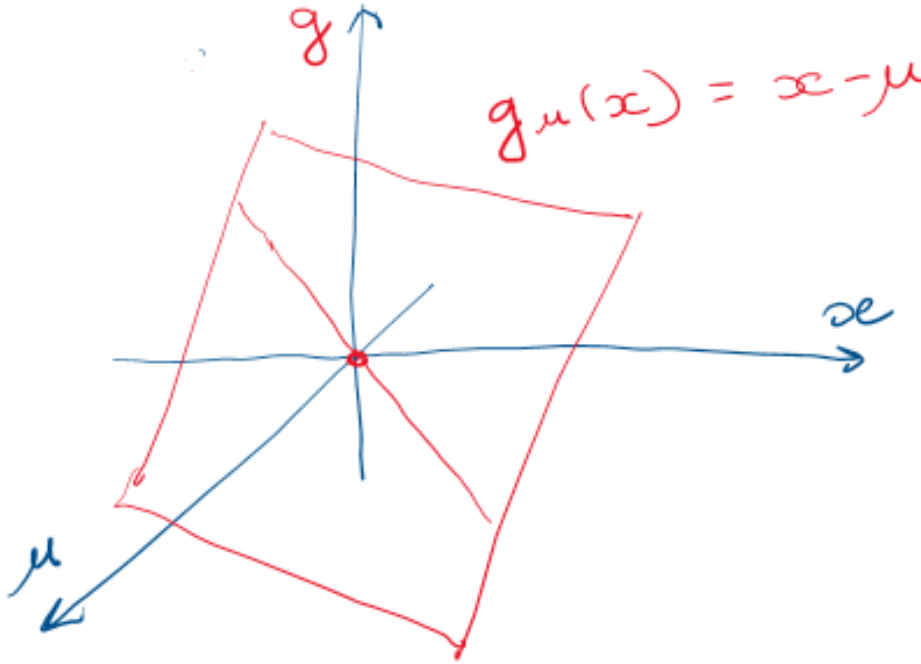


Figure 6: Boris's diagram 4: $g_\mu(x) = x - \mu$ as a 3D surface over the (x, μ) -plane. It is a tilted plane (not a curved surface), rising along x and falling along μ .

The substitution $y = x - \mu$ changes variables in the likelihood surface:

$$L(x, \mu) = \phi(x - \mu) = \phi(y).$$

In the new coordinate y , the likelihood is simply $\phi(y)$ — a function of y alone, with *no μ anywhere*.

Green curtains: horizontal after the transform

Take any fixed true value, say $\mu_t = 5$, and sample $x \in \{4, 4.5, 5.1\}$. Under the pivot map at $\mu = \mu_t$:

$$y_i = x_i - 5 \in \{-1, -0.5, 0.1\}.$$

Each transformed value y_i gives a “green curtain” on the (y, μ) -surface: the function $\mu \mapsto \phi(y_i)$. But $\phi(y_i)$ does not depend on μ — it is a constant as μ varies. The curtain is a **horizontal plane**: the same height everywhere along the μ -axis.

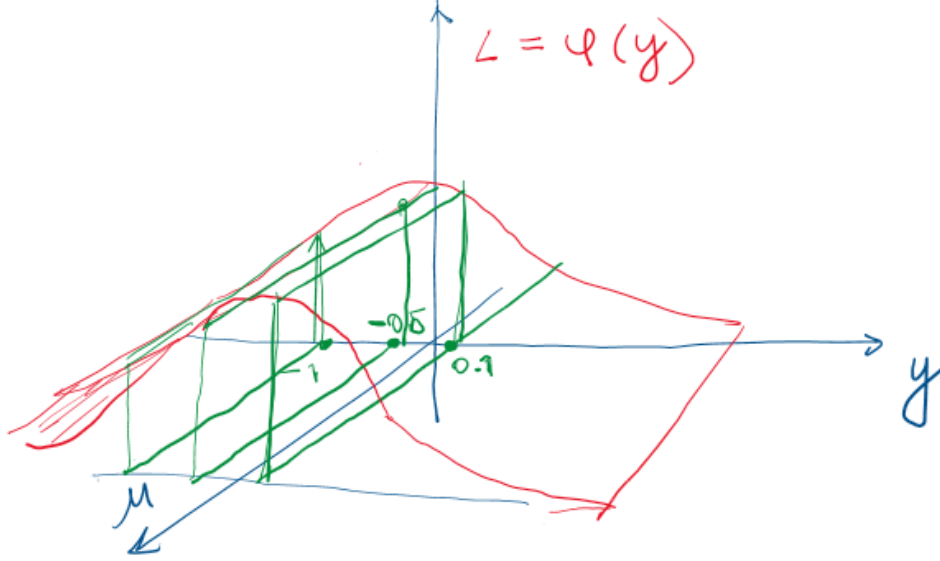


Figure 7: Boris’s diagram 5: the surface $\phi(y)$ over the (y, μ) -plane. The green curtains at fixed values of y are horizontal — $\phi(y)$ does not depend on μ . Compare with diagram 2: the orange curtains were bell-shaped; the green curtains are flat.

Remark 16 (The visual statement of the pivot property). In the original (x, μ) space, the orange curtain at observation x_i is a bell curve in μ — it rises, peaks at $\mu = x_i$, and falls. Picking a different μ gives different likelihood values.

After the pivot transform, the green curtain at $y_i = x_i - \mu_t$ is a horizontal line — the same value $\phi(y_i)$ regardless of μ . This is the geometric meaning of “the distribution of the pivot does not depend on the parameter”: the likelihood (as a function of the abstract μ) is flat.

Boris captures the key paradox: *the result does not depend on μ , but the transform did depend on μ* . Specifically, the $y_i = x_i - \mu_t$ were computed using the true $\mu_t = 5$. If we had used a different μ_t , we would have gotten different y_i ’s — the *values* would be different. Yet for any fixed y_i , the function $\mu \mapsto \phi(y_i)$ is flat. The transform carries the μ -dependence into the *value* of y_i ; the standard normal density ϕ then displays that value without any further μ -dependence.

This is the same observation from note 60, now seen geometrically: fixing ω (the randomness) fixes x , which fixes $y = x - \mu_t$ at a particular value; the flat curtain at that value is the horizontal section of ϕ . The distribution of y over random ω is $\mathcal{N}(0, 1)$ regardless of μ_t — but which value y takes does depend on μ_t .

1.7 Regular conditional distribution and the conditional likelihood of $Y \mid X$

We follow Boris’s notes of July 17, 2026 (note 65).

The conditioning problem

The model specifies the conditional distribution of U given X :

$$U \mid X = x \sim \mathcal{N}(0, \sigma^2(x)), \quad \sigma^2(x) = h(\alpha_0 + \alpha_1 x).$$

For continuous X , the event $\{X = x\}$ has probability zero. The conditional probability $P(U \in A \mid X = x)$ cannot be defined as a ratio $P(U \in A, X = x)/P(X = x)$ — the denominator is zero. For each individual x one can find a version of the conditional probability defined up to a P_X -null set. The difficulty: *there is no single null set that works for all x simultaneously* in general.

Regular conditional distribution

For abstract probability spaces this is a genuine obstruction, and counterexamples exist. However, for Polish spaces such as $\mathbb{R}$ or $\mathbb{R}^d$, a canonical simultaneous version always exists.

Definition 17. A **regular conditional distribution** of U given X is a function

$$\kappa : \Omega \times \mathcal{B}(\mathbb{R}) \longrightarrow [0, 1]$$

satisfying:

1. For every fixed $\omega \in \Omega$: the map $A \mapsto \kappa(\omega, A)$ is a probability measure on $\mathbb{R}$.
2. For every fixed Borel set $A \in \mathcal{B}(\mathbb{R})$: the map $\omega \mapsto \kappa(\omega, A)$ is $\sigma(X)$ -measurable.
3. $\kappa(\omega, A) = P(U \in A \mid X)(\omega)$ P -almost surely.

The map $\omega \mapsto \kappa(\omega, \cdot)$ is a *random measure* — it assigns a probability measure on $\mathbb{R}$ to each sample point ω . Condition (1) ensures it is a proper measure for each ω ; condition (2) ensures it is measurable.

In our model, writing $x = X(\omega)$, the regular conditional distribution takes the specific form

$$\kappa(\omega, A) = \mathcal{N}(0, h(\alpha_0 + \alpha_1 x))(A) = \int_A \frac{1}{\sigma_x} \phi\left(\frac{u}{\sigma_x}\right) du,$$

where $\sigma_x = \sqrt{h(\alpha_0 + \alpha_1 x)}$.

Conditional distribution of Y given X : non-centration

Switching from U to $Y = \beta_0 + \beta_1 X + U$ shifts the conditional distribution. Where $U \mid X = x$ is centred at zero, $Y \mid X = x$ is centred at the regression mean:

$$Y \mid X = x \sim \mathcal{N}(\beta_0 + \beta_1 x, \sigma^2(x)).$$

The shift $\beta_0 + \beta_1 x$ is the “non-centration” Boris refers to: $Y \mid X$ has the same spread as $U \mid X$ but is displaced. The regular conditional distribution of Y given X is thus

$$\kappa_Y(\omega, A) = \mathcal{N}(\beta_0 + \beta_1 x, h(\alpha_0 + \alpha_1 x))(A), \quad x = X(\omega).$$

Conditional likelihood and log-likelihood

The conditional density of Y_i given $X_i = x_i$ is

$$f_{Y|X}(y_i \mid x_i; \beta, \alpha) = \frac{1}{\sigma_{x_i}} \phi\left(\frac{y_i - \beta_0 - \beta_1 x_i}{\sigma_{x_i}}\right),$$

the density of the regular conditional distribution κ_Y at $x = x_i$. For n independent pairs, the conditional likelihood is

$$L(\beta, \alpha; y_{1:n} \mid x_{1:n}) = \prod_{i=1}^n \frac{1}{\sigma_{x_i}} \phi\left(\frac{y_i - \beta_0 - \beta_1 x_i}{\sigma_{x_i}}\right).$$

The log-likelihood is

$$l(\beta, \alpha) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \sum_{i=1}^n \log \sigma_{x_i}^2 - \frac{1}{2} \sum_{i=1}^n \frac{(y_i - \beta_0 - \beta_1 x_i)^2}{\sigma_{x_i}^2}. \quad (1)$$

This is the BP log-likelihood, now derived with full measure-theoretic justification.

Remark 18 (Three derivations of the same formula). Equation (1) has now been derived three ways: (§ 1.1, note 58) via the change-of-variables Jacobian; (§ 1.2, note 59) via plugging the pivot into the joint normal density; and here via the density of the regular conditional distribution. All three are equivalent; the present route is the most rigorous, identifying $f_{Y|X}(y|x; \beta, \alpha)$ as the Radon-Nikodym density of $\kappa_Y(\omega, \cdot)$ with respect to Lebesgue measure.

1.8 Mixed partial derivatives: geometric meaning and the block-diagonal assumption

Stationarity does not require zero mixed partials

Boris conjectured that the block-diagonal condition on the Fisher information matrix might follow from the stationarity of the restricted MLE. This is not correct. A stationary point requires the *gradient* to vanish; the mixed partial derivatives of a function are a separate property of the Hessian.

Consider two functions of two real variables.

Function A (decoupled):

$$f(x, y) = x^2 + 3y^2.$$

Gradient: $\nabla f = (2x, 6y)^\top$. Stationarity at $(0, 0)$: $\nabla f = 0$. Hessian:

$$H_f = \begin{pmatrix} 2 & 0 \\ 0 & 6 \end{pmatrix}.$$

Mixed partial: $f_{xy} = 0$. The Hessian is diagonal.

Function B (coupled):

$$g(x, y) = x^2 + 2xy + 3y^2 = (x + y)^2 + 2y^2.$$

Gradient: $\nabla g = (2x + 2y, 2x + 6y)^\top$. Setting to zero: $x + y = 0$ and $x + 3y = 0$, giving the unique stationary point $(0, 0)$. Hessian:

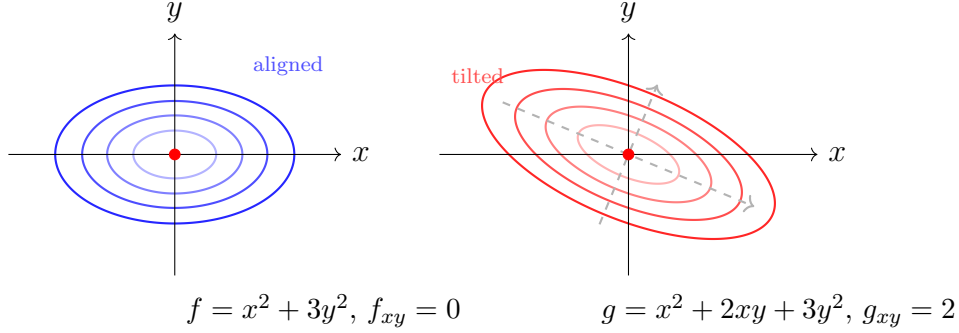
$$H_g = \begin{pmatrix} 2 & 2 \\ 2 & 6 \end{pmatrix}.$$

Mixed partial: $g_{xy} = 2 \neq 0$. The Hessian is *not* diagonal, even though $(0, 0)$ is a stationarity point for both f and g .

Both functions have the same stationary point and both are positive definite at that point ($\det H_f = 12 > 0$, $\det H_g = 8 > 0$, traces positive). Yet $f_{xy} = 0$ while $g_{xy} = 2$. **Stationarity tells us nothing about mixed partials.**

Geometric meaning: aligned vs. tilted level curves

The level curves $\{(x, y) : f(x, y) = c\}$ of the two functions reveal the difference visually.



For f (left): the level curves are ellipses with semi-axes aligned along the x - and y -directions. The curvature in x does not interact with the curvature in y .

For g (right): the level curves are tilted ellipses. The dashed lines show the eigenvectors of H_g , which are the true axes of curvature — they do not coincide with the coordinate axes. The mixed partial $g_{xy} = 2$ is the coefficient describing this tilt: how much the slope in x changes when we move in y .

What $f_{xy} = 0$ means in words

The mixed partial $f_{xy}(x_0, y_0)$ answers: if we increase y slightly from y_0 , does the slope of f in the x -direction change? If $f_{xy} = 0$ everywhere, the answer is no: the x -profile of f (a cross-section by fixed y) has the same shape regardless of y . The two variables are *curvature-independent*: knowing the curvature structure in x tells you nothing about how y enters.

Equivalently: the Hessian is diagonal iff there exists a decomposition $f(x, y) = u(x) + v(y)$ (locally near the stationary point, up to second order).

The Fisher information block-diagonal assumption

In the BP model, the Fisher information matrix is

$$\mathcal{J}(\theta) = -E \left[\frac{\partial^2 l}{\partial \theta \partial \theta'} \right], \quad \theta = (\alpha, \beta),$$

and the block-diagonal assumption is $\mathcal{J}_{\alpha\beta} = 0$.

Remark 19 (Is this an assumption or a theorem?). For the specific BP log-likelihood it is a *theorem*, not an extra assumption. We verified it earlier by direct computation: the cross second derivative $\partial^2 l / \partial \alpha_j \partial \beta_k$ contains factors u_t or $(u_t^2 - \sigma_t^2)$, whose expectation under the model is zero. So $\mathcal{J}_{\alpha\beta} = 0$ follows from $E[u_t] = 0$ and $E[u_t^2] = \sigma_t^2$.

In Rao's general framework (Section 6e.3), the block-diagonal condition is stated as a *hypothesis* on the model, because Rao is not working with a specific parametric family. For any particular model one must verify it. BP do so implicitly, and Rao provides the general theorem that, given this condition, the LM statistic takes the simplified form $\text{LM} = \hat{d}_2' \hat{\mathcal{J}}_{22}^{-1} \hat{d}_2$.

Geometrically: $\mathcal{J}_{\alpha\beta} = 0$ means the “likelihood bowl” near the restricted MLE has axes aligned with the (α, β) coordinate system. Optimising over β for fixed α is decoupled from optimising over α for fixed β . It is precisely this decoupling that allows the nuisance β to be eliminated without affecting the α -part of the test statistic.

1.9 Rao's asymptotic score result: the theoretical basis for χ^2

Setup (Rao 1973, Section 6e.1)

Let $x_1, \dots, x_n$ be i.i.d. with density $p(x, \theta)$ and $\theta = (\theta_1, \dots, \theta_q)^\top \in \mathbb{R}^q$. Define the log-likelihood, score vector, and Fisher information:

$$\begin{aligned} l(\theta) &= \sum_{i=1}^n \log p(x_i, \theta), \\ V(\theta) &= \frac{\partial l}{\partial \theta} \quad (q\text{-vector}), \\ F(\theta) &= -E \left[\frac{\partial^2 l}{\partial \theta \partial \theta'} \right] = n \mathcal{I}(\theta) \quad (q \times q \text{ matrix}). \end{aligned}$$

By the score identity: $E_\theta[V(\theta)] = 0$. By the CLT applied to i.i.d. score contributions:

$$\frac{1}{\sqrt{n}} V(\theta) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\theta)) \quad \text{as } n \rightarrow \infty.$$

Simple hypothesis: $H_0 : \theta = \theta_0$ (Section 6e.2)

Rao proposes the **efficient score statistic**:

$$S_0 = V_0' F_0^{-1} V_0, \quad V_0 = V(\theta_0), \quad F_0 = F(\theta_0).$$

Under H_0 : $V_0/\sqrt{n} \rightarrow \mathcal{N}(0, \mathcal{I})$, so $V_0' F_0^{-1} V_0 \xrightarrow{d} \chi^2(q)$.

This is asymptotically equivalent to the likelihood-ratio statistic $-2 \log \Lambda$ and Wald's statistic W_0 . All three have the same limiting $\chi^2(q)$ distribution under H_0 (proved by Rao via Taylor expansion and the properties of efficient estimators).

The advantage of S_0 : it does not require computing the MLE $\hat{\theta}$ explicitly. It only requires evaluating the log-likelihood at θ_0 .

Composite hypothesis (Section 6e.3, pp. 418–419)

Now the null is $H_0 : R_j(\theta) = 0$ for $j = 1, \dots, k$ — k restrictions on q parameters. The free parameters under H_0 number $s = q - k$.

Let $\hat{\theta}$ be the unrestricted MLE and $\hat{\theta}^*$ the *restricted MLE* (constrained to H_0). Rao's efficient score test for the composite hypothesis (equation 6e.3.6) is:

$$S_* = V_*' F_*^{-1} V_*, \quad V_* = V(\hat{\theta}^*), \quad F_* = F(\hat{\theta}^*).$$

Proposition 20 (Rao 1973, p. 419). *Under H_0 and regularity conditions, $S_* \xrightarrow{d} \chi^2(k)$. All three statistics S_* , W_* , and $-2 \log \Lambda_*$ are asymptotically equivalent and have this limit.*

Why the restricted MLE and not the unconstrained one

At the unconstrained MLE $\hat{\theta}$: $V(\hat{\theta}) = 0$ by the first-order conditions of optimisation. Therefore $S_* = 0$ — identically zero and useless.

At the restricted MLE $\hat{\theta}^*$: the free parameters $(\theta_3, \dots, \theta_q)$ are at their optimum under the constraint, so their score components are zero. But the score in the constrained directions $(\theta_1, \dots, \theta_k)$ is generally non-zero and measures how strongly the likelihood “pushes” against the constraint. It is this non-zero part that constitutes the test.

Why the result is $\chi^2(k)$, not $\chi^2(q)$

Under H_0 , $\hat{\theta}^* \xrightarrow{P} \theta_0$ (the restricted MLE is consistent for the true parameter). So $V_* \approx V(\theta_0)$ for large n .

The score vector $V(\theta_0)$ is a q -dimensional random vector with mean 0. But k of its coordinates are eliminated by the first-order conditions at $\hat{\theta}^*$ (the unconstrained directions have zero score). Only the k constrained coordinates remain, and under the block-diagonal assumption these are asymptotically $\mathcal{N}(0, F_{22})$ in k dimensions. The quadratic form $V_*' F_*^{-1} V_*$ then equals approximately $\chi^2(k)$.

Formally: Rao proves (p. 419) that $2[l(\hat{\theta}) - l(\hat{\theta}^*)] \approx V_*'(F^{-1} - MH^{-1}M')V_*$ where M is the matrix of derivative of the constraints with respect to the free parameters, and H is their information matrix. The idempotency of the relevant matrix gives χ^2 with degrees of freedom equal to the rank of the idempotent, which is $k = q - s$.

Simple example: $\mathcal{N}(\mu, 1)$, $H_0 : \mu = 0$

n observations, $\theta = \mu$, $q = 1$, $k = 1$. The null is simple. H_0 constrains $\mu = 0$, leaving no free parameters.

$$l(\mu) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \sum (x_i - \mu)^2, \quad V = \sum (x_i - \mu), \quad F = n.$$

Restricted MLE: $\hat{\mu}^* = 0$ (the only value satisfying H_0).

Score at restricted MLE: $V_* = \sum x_i = n\bar{x}$.

Rao statistic: $S_* = (n\bar{x})^2/n = n\bar{x}^2$.

Under H_0 : $\sqrt{n}\bar{x} \sim \mathcal{N}(0, 1)$, so $n\bar{x}^2 \sim \chi^2(1)$. ✓ This is the squared z -test.

Example with nuisance: $\mathcal{N}(\mu, \sigma^2)$, $H_0 : \mu = 0$, σ^2 unknown

Now $\theta = (\mu, \sigma^2)$, $q = 2$, $k = 1$, $s = 1$ (σ^2 free).

Restricted MLE: $\hat{\mu}^* = 0$, $\hat{\sigma}^{*2} = n^{-1} \sum x_i^2$.

Score at $\hat{\theta}^*$:

$$V_\mu|_{\hat{\theta}^*} = \frac{\sum x_i}{\hat{\sigma}^{*2}} = \frac{n\bar{x}}{\hat{\sigma}^{*2}}, \quad V_{\sigma^2}|_{\hat{\theta}^*} = -\frac{n}{2\hat{\sigma}^{*2}} + \frac{\sum x_i^2}{2\hat{\sigma}^{*4}} = 0 \quad \checkmark$$

(the σ^2 -score is zero by the definition of $\hat{\sigma}^{*2}$).

Block-diagonal Fisher information for $\mathcal{N}(\mu, \sigma^2)$: $\mathcal{J}_{\mu, \sigma^2} = 0$ (a standard calculation). Therefore:

$$S_* = \hat{V}_\mu^2 / F_{\mu\mu} = \frac{(n\bar{x}/\hat{\sigma}^{*2})^2}{n/\hat{\sigma}^{*2}} = \frac{n\bar{x}^2}{\hat{\sigma}^{*2}} \xrightarrow{d} \chi^2(1) \text{ under } H_0.$$

The nuisance σ^2 entered only through $\hat{\sigma}^{*2}$ and dropped out cleanly, thanks to the block-diagonal structure (§ 1.8).

Connection to BP

BP set $\theta = (\alpha_0, \alpha_1, \dots, \alpha_q, \beta_0, \beta_1, \dots, \beta_p)$ with $H_0 : \alpha_1 = \dots = \alpha_q = 0$. The number of restrictions is $k = q$. Restricted MLE = OLS. Block-diagonal Fisher information = verified from $E[u_t] = 0$.

Rao's theorem then gives: $S_* \xrightarrow{d} \chi^2(q)$ under H_0 . This is the theoretical basis for the $\chi^2(p-1)$ in BP's notation (where $p = q + 1$ includes the intercept in α).

1.10 The LM statistic: conditional score, restricted MLE, and block structure

We follow Boris's notes of July 18, 2026 (note 66).

What BP call “the score”

In the paper, BP differentiate the *conditional* log-likelihood $l(\theta) = \sum_i \log f_{Y|X}(y_i | x_i; \theta)$ with respect to θ and call the result the score. This is the standard score vector:

$$d(\theta) = \frac{\partial l(\theta)}{\partial \theta}.$$

BP refer to Rao (1973), *Linear Statistical Inference and Its Applications*, pp. 418–419, for the general Lagrange multiplier framework.¹ The Probabilist could not locate it freely online; Boris found both a paid and a free version. We will consult it later.

The general LM statistic

From Rao (via BP): let $l(\theta)$ be the log-likelihood with score $d = \partial l / \partial \theta$ and Fisher information $\mathcal{J} = -E[\partial^2 l / \partial \theta \partial \theta']$. The LM statistic for testing $H_0 : \phi(\theta) = 0$ is

$$\text{LM} = \hat{d}' \hat{\mathcal{J}}^{-1} \hat{d},$$

where hats denote evaluation at $\hat{\theta}$ — the **restricted** MLE satisfying $\phi(\hat{\theta}) = 0$.

The restricted MLE

Boris emphasises: $\hat{\theta}$ is the argmax of the conditional likelihood *subject to the constraint* imposed by H_0 , not the unconstrained argmax. In our concrete setting ($p = q = 1$):

$$\theta = (\alpha_0, \alpha_1, \beta_0, \beta_1) \in \mathbb{R}^4.$$

The constraint is $\phi(\theta) = \alpha_1 = 0$. The restricted MLE is

$$\hat{\theta} = \underset{\alpha_0, \alpha_1=0, \beta_0, \beta_1}{\operatorname{argmax}} l(\alpha_0, 0, \beta_0, \beta_1),$$

a three-dimensional optimisation. One can show that $\hat{\beta}$ coincides with the OLS estimator and $\hat{\alpha}_0$ is determined by $\hat{\sigma}^2 = n^{-1} \sum_i \hat{u}_i^2$.

How the score splits at $\hat{\theta}$

Partition $\theta = (\alpha, \beta)$ and correspondingly $d = (d_\alpha, d_\beta)^\top$. At the restricted MLE, the likelihood is stationary in the unconstrained directions β :

$$\hat{d}_\beta = \left. \frac{\partial l}{\partial \beta} \right|_{\hat{\theta}} = 0.$$

This is just the first-order condition: $\hat{\theta}$ is a genuine interior optimum in β , so the gradient vanishes there.

The constrained direction α_1 is fixed at zero by H_0 , so $\hat{d}_{\alpha_1}$ is generally *non-zero* — it measures how strongly the likelihood wants to move in the α_1 direction, i.e. how much evidence there is against H_0 .

¹Rao, C.R. (1973). *Linear Statistical Inference and Its Applications*, 2nd ed. Wiley. A digitised copy: <https://ia601508.us.archive.org/25/items/in.ernet.dli.2015.40834/2015.40834.Linear-Statistical-Inference-And-Its-Applications.pdf>.

Thus the score at $\hat{\theta}$ takes the form

$$\hat{d} = \begin{pmatrix} \hat{d}_{\alpha_0} \\ \hat{d}_{\alpha_1} \\ 0 \\ 0 \end{pmatrix},$$

where the bottom two zeros are the (β_0, β_1) components.

Block-diagonal assumption on $\mathcal{J}$

BP additionally assume that the Fisher information matrix is block-diagonal between α and β :

$$\mathcal{J}_{\alpha\beta} := -E \left[\frac{\partial^2 l}{\partial \alpha \partial \beta'} \right] = 0.$$

Boris notes the distinction: $\hat{d}_\beta = 0$ is a consequence of the first-order conditions (a fact about the optimum). The vanishing of $\mathcal{J}_{\alpha\beta}$ is an *additional assumption* that BP impose — and that we verified earlier by direct computation: the cross-derivative has expectation zero because $E[u_i] = 0$.

Under block-diagonality, the general LM statistic reduces to

$$\boxed{\text{LM} = \hat{d}'_\alpha \hat{\mathcal{J}}_{\alpha\alpha}^{-1} \hat{d}_\alpha},$$

where only the α -block appears. The β -coordinates are zero and contribute nothing.

Remark 21 (What remains). We have the general form. The next subsection evaluates $\hat{d}_\alpha$ and $\hat{\mathcal{J}}_{\alpha\alpha}$ for the specific BP log-likelihood and shows the result equals $\frac{1}{2} \widehat{\text{ESS}}$ from the auxiliary regression of $g_i = \hat{u}_i^2 / \hat{\sigma}^2$ on z_i .

1.11 The Breusch–Pagan test as a score test

We follow the Probabilist's notes of June 7, 2026.

Model hierarchy

Simple regression with normally distributed errors:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \sim \text{ind } \mathcal{N}(0, \sigma_i^2), \quad (1)$$

$$\sigma_i^2 = h(\alpha_0 + \alpha_1 z_i), \quad (2)$$

where $h(\cdot)$ is any twice-differentiable positive function, and z_i is an observable variable (often $z_i = x_i$, but not necessarily).

Model (1)&(2) is a *sub-model* of model (1): it adds the parametric constraint (2) on how the variance moves. The null hypothesis of homoscedasticity is

$$H_0 : \alpha_1 = 0.$$

Under H_0 , equation (2) gives $\sigma_i^2 = h(\alpha_0) =: \sigma^2$ for all i — a constant.

The model-space picture

The ideal test would compare the heteroscedastic part of model (1) (white) against the homoscedastic part (yellow). We cannot do this directly — the white region has no finite-dimensional parametrisation. Instead, the BP test embeds the problem into the parametric sub-model (1)&(2) and tests the green region against the yellow one, i.e. $\alpha_1 = 0$ inside (2).

Note the Probabilist's point: the z_i are *specific* and must be observable. The authors kept z_i general (rather than always setting $z_i = x_i$) to nest heteroscedastic models from earlier literature.

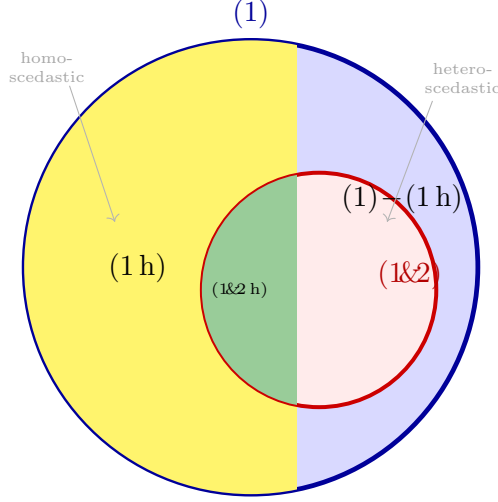


Figure 8: The Probabilist's diagram (June 7, 2026). The large dark-blue circle is the full model (1) — simple regression with unrestricted heteroscedasticity. It splits into the yellow region (1 h) (homoscedastic) and the white region (1) – (1 h) (heteroscedastic). The smaller red-contour circle (1&2) is the sub-model that additionally imposes the parametric form (2) on the variance; it splits into the green region (1&2 h) (homoscedastic sub-model) and its local complement. We want to test white against yellow, but instead test blue against yellow — the difference is parametrised by $\alpha_1 = 0$ or not.

Why the score test

The score (LM) test is ideally suited here. Under H_0 the restricted MLE is just OLS — no iterative optimisation needed. The test asks: does the log-likelihood *want to move* away from the null, i.e. is its slope with respect to α_1 large at the OLS solution?

Log-likelihood and score

The log-likelihood for model (1)&(2) is

$$\ell(\beta, \alpha) = \text{const} - \frac{1}{2} \sum_i \log \sigma_i^2 - \frac{1}{2} \sum_i \frac{\varepsilon_i^2}{\sigma_i^2}, \quad \sigma_i^2 = h(\alpha_0 + \alpha_1 z_i).$$

Differentiating with respect to α_1 and evaluating at H_0 (where $\sigma_i^2 = \hat{\sigma}^2$ and $\varepsilon_i = \hat{u}_i$):

$$\left. \frac{\partial \ell}{\partial \alpha_1} \right|_{H_0} = \frac{h'(\hat{\alpha}_0)}{2\hat{\sigma}^2} \sum_i \left(\frac{\hat{u}_i^2}{\hat{\sigma}^2} - 1 \right) z_i.$$

Define $g_i := \hat{\sigma}^{-2} \hat{u}_i^2$ and $f_i := g_i - 1$. The score is proportional to $\sum_i f_i z_i$ — the sample covariance between the normalised squared residuals and z_i .

Fisher information and the cancellation of h'

The (α_1, α_1) block of the Fisher information, evaluated at H_0 , is

$$\mathcal{J}_{\alpha_1 \alpha_1} \big|_{H_0} = \frac{[h'(\hat{\alpha}_0)]^2}{2\hat{\sigma}^4} \sum_i z_i^2.$$

The LM statistic $\hat{d}'\hat{\mathcal{J}}^{-1}\hat{d}$ therefore equals

$$LM = \frac{(\sum_i f_i z_i)^2}{\sum_i z_i^2},$$

where the factor $h'(\hat{\alpha}_0)$ has *cancelled completely*. The statistic does not depend on the choice of h — only on the residuals $\hat{u}_i$ and the variable z_i .

Connection to the auxiliary regression

Write $z_i = (1, z_i)'$ for the general case with a constant in (2). Breusch and Pagan show (by the algebra $f = g - \iota$, $\iota'g = N$, $f'Z(Z'Z)^{-1}Z'\iota = 0$) that the LM statistic equals one half the explained sum of squares in the regression of g_i on z_i :

$$LM = \frac{1}{2} \text{ESS}(g_i \sim z_i) \xrightarrow{d} \chi_{p-1}^2 \quad \text{under } H_0,$$

where $p - 1$ is the number of variance regressors (excluding the constant α_0). For simple regression with $z_i = x_i$ this gives χ_1^2 .

The practical formula $nR^2 \sim \chi_1^2$ follows because $\text{ESS} = R^2 \cdot \text{TSS}$ and TSS of g_i equals $2n$ asymptotically under normality (since $g_i = \hat{\sigma}^{-2}\hat{u}_i^2 \approx \chi_1^2$ and $\text{Var}(\chi_1^2) = 2$).

Remark (finite samples). The χ^2 approximation can be poor for small n . Breusch and Pagan's own simulations (Table I of the paper) show that at $n = 20$ the actual Type I error at a nominal 5% level is closer to 1–2%, so conservative critical values are advisable. The Koenker (“studentised”) variant drops the normality assumption and is better calibrated in small samples.